7. (a) Skiers and mountaineers use hand warmers in very cold weather conditions. Hand warmers are small packets that produce heat when needed. Some ski clothing is specially designed to hold a hand warmer.

Table **1** describes how three types of hand warmers work.

A Air-activated (disposable)	**B** Battery powered (reusable)	**C** Supersaturated solution (reusable)
The packaging seal is broken allowing air to reach the chemicals causing a chemical reaction to occur. This type of hand warmer can only be used once. 	A metal coil heats up when the device is switched on. The device needs recharging. 	A metal button on the packaging is pressed. This causes crystals to form in the solution. The hand warmer can be reactivated by placing it in boiling water. 

Table 1

Examiner only

Table **2** shows some information about each type of hand warmer.

Type of hand warmer	Cost (£)	Time to warm up after activation	Temperature after being activated and placed in a freezer (°C)				
			after 15 mins	after 30 mins	after 60 mins	after 90 mins	after 120 mins
A	1	less than 1 min	39	39	38	38	37
B	80	less than 1 min	32	29	28	27	26
C	24	less than 1 min	42	34	27	18	8

Table 2

(i) Tick (✓) the box next to the correct statement. [1]

Gloves need to be worn when using hand warmers ☐

Boiling water is used to recharge battery powered hand warmers ☐

Some chemical reactions give out heat energy ☐

All hand warmers are reusable ☐

(ii) Give **two** reasons why hand warmer **A** is the most popular choice. [2]

Reason 1 ....

....

Reason 2 ....

....

(b) An air-activated hand warmer contains several chemicals mixed together. One of these chemicals is iron.

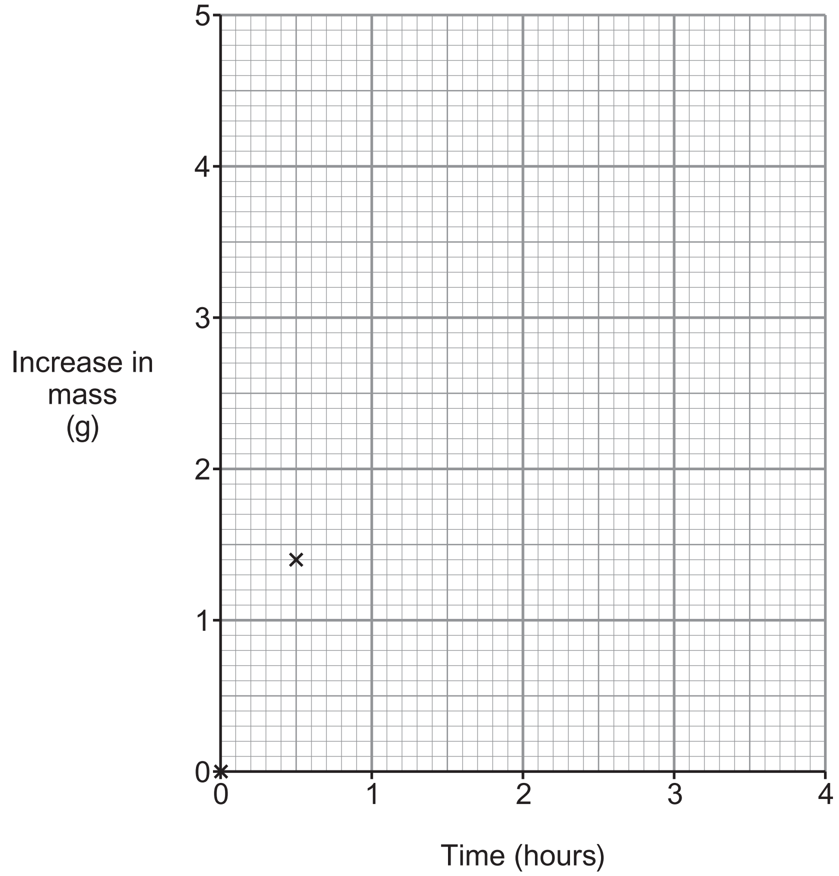
When the pack is opened, a reaction takes place between iron and oxygen in the air, causing an increase in mass.

A student investigated the increase in mass of the opened pack over several hours. The results are shown below.

Time (hours)	Increase in mass (g)
0.0	0.0
0.5	1.4
1.0	2.6
1.5	3.7
2.0	4.4
2.5	4.7
3.0	4.8
3.5	4.8
4.0	4.8

Examiner only

(i) Plot the mass increase of the pack against time on the grid and draw a suitable line. Two plots have been done for you. [3]



(ii) Tick (✓) the box next to the time it takes for the reaction to finish. [1]

2 hours ☐ 3 hours ☐ 4 hours ☐ 5 hours ☐

(iii) Tick (✓) the box next to the statement that best explains the shape of the graph. [1]

Iron reacts with oxygen forming iron oxide until all the oxygen is used up	☐
Heat formed expands the iron	☐
Iron oxide loses oxygen, forming iron	☐
Iron reacts with oxygen forming iron oxide until all the iron is used up	☐

8